Lab 5 – Network Visualization with Gephi

In this lab, you will get hands-on experience with Gephi, learning how to import data, visualize networks, and perform basic analysis using its core features.

1. Introduction to Gephi

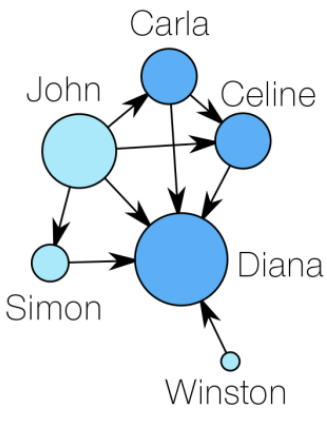
Gephi is a powerful, open-source tool for exploring and visualizing network (graph) data. It allows you to interactively manipulate network structures, adjust visual properties, and reveal hidden patterns. Gephi is available for Windows, macOS, and Linux.

For installation, please follow the **official website & installation guide:** Gephi Installation Instructions

2. What is Social Network Analysis?

A network consists of:

- **Nodes (vertices):** The entities in the network (e.g., people, organizations).
- **Edges (links):** The relationships or interactions between nodes.



Nodes

Id,Label,Attribute
1,John,1
2,Carla,2
3,Simon,1
4,Celine,2
5,Winston,1
6,Diana,2

Edges

Source,Target
1,2
1,3
1,4
1,6
2,4
2,6
3,6
4,6
5,6

Each node can have attributes (e.g., name, gender, value), and each edge connects two nodes (source and target). In social network analysis, we often use node size, color, and layout to represent different properties and reveal insights.

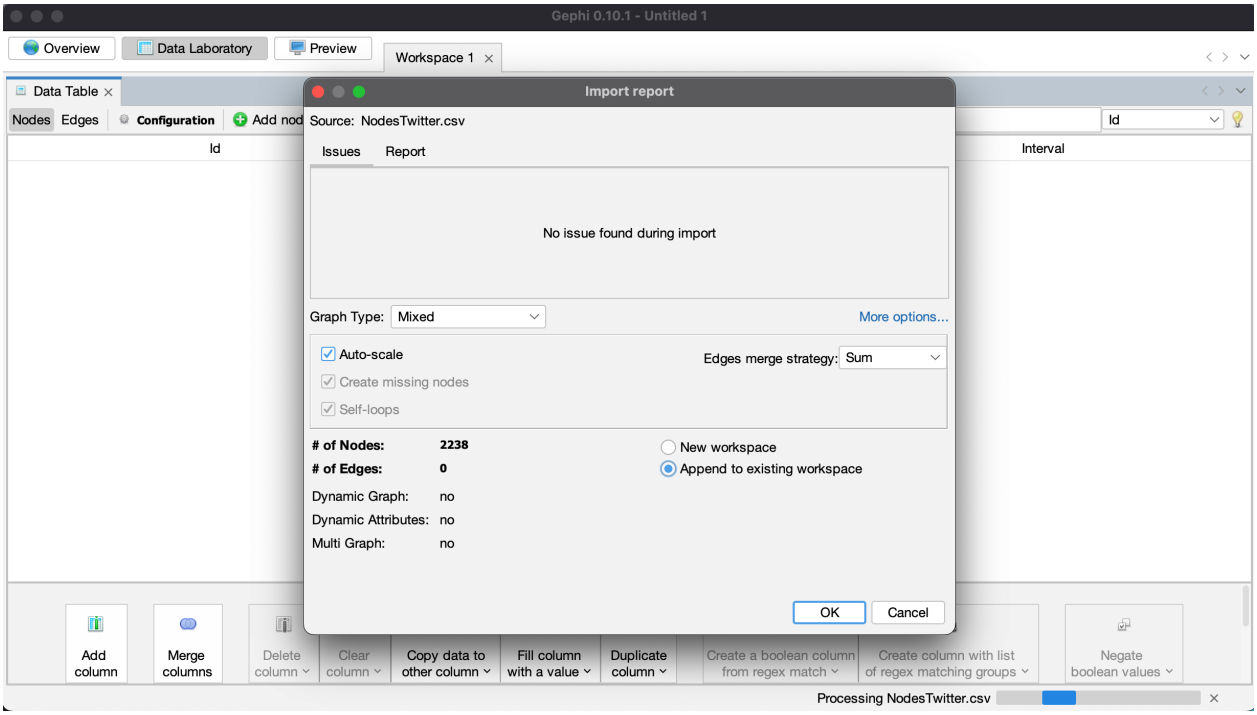
3. Dataset Overview

You will use two files (download from Moodle):

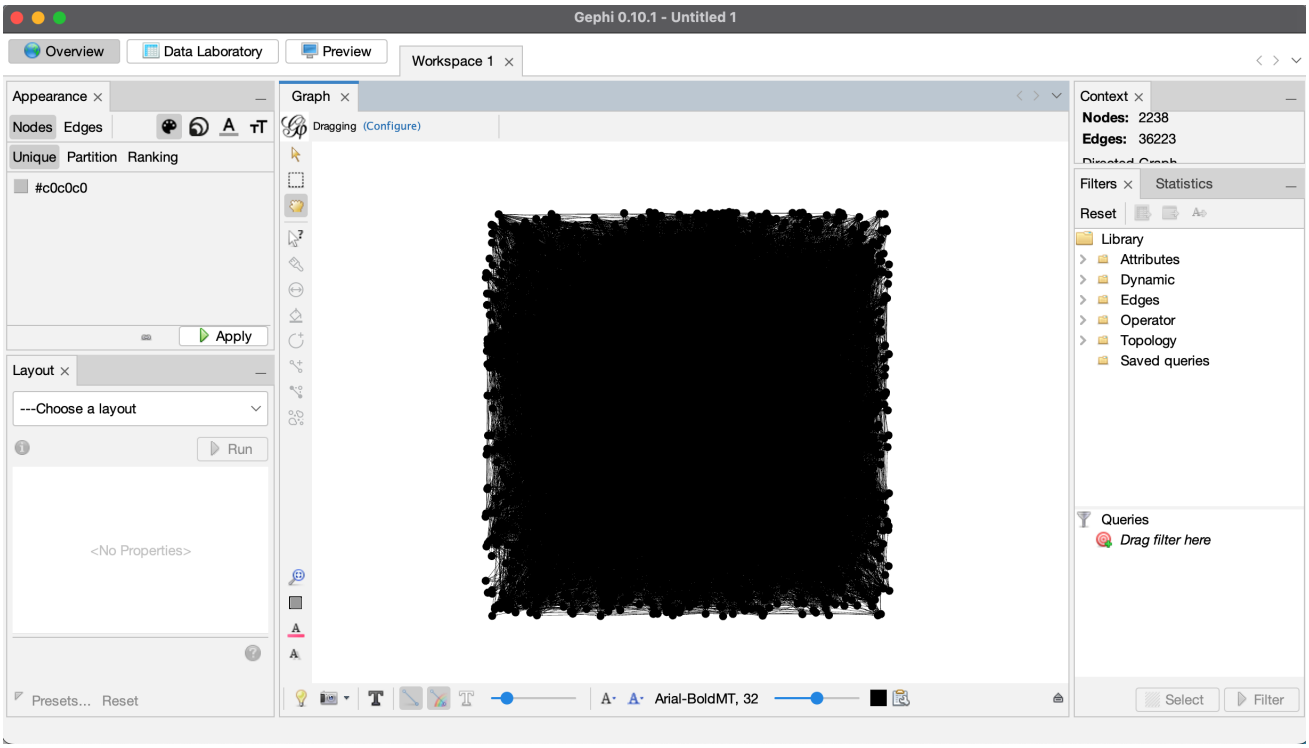
- **NodesTwitter.csv:** Contains node IDs, labels, a 'sex' attribute, and a random value.
- **EdgesTwitter.csv:** Contains pairs of node IDs representing connections (edges).

4. Importing Data into Gephi

1. **Start Gephi and create a new project.**
2. Go to the **Data Laboratory** tab.
3. Click **Import Spreadsheet** and select `NodesTwitter.csv`.
   - Use the default settings and click **Finish**.
   - Select **"Append to existing workspace"** to ensure your data is added to the current project.



4. Next, import similarly for `EdgesTwitter.csv`.
5. After importing, click **Overview** to see the initial network graph. The layout will be random at first.



5. Basic Visualization Steps

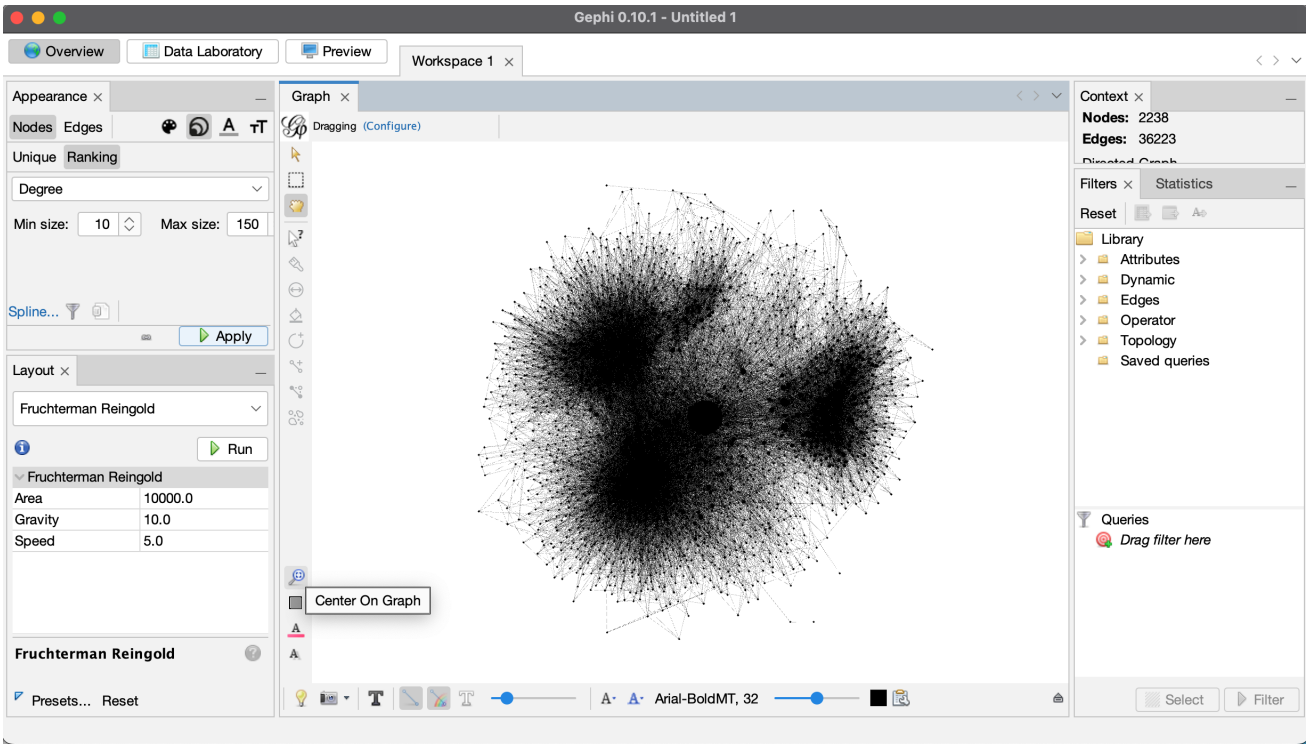
a. Adjust Node Size by Degree

- In the **Appearance** panel (top left), select **Nodes** and the **Ranking** option.
- Choose **Degree** as the ranking attribute.
- Set min and max size (e.g., 10–150).
- Click **Apply** to update node sizes based on their number of connections.

b. Improve Graph Layout

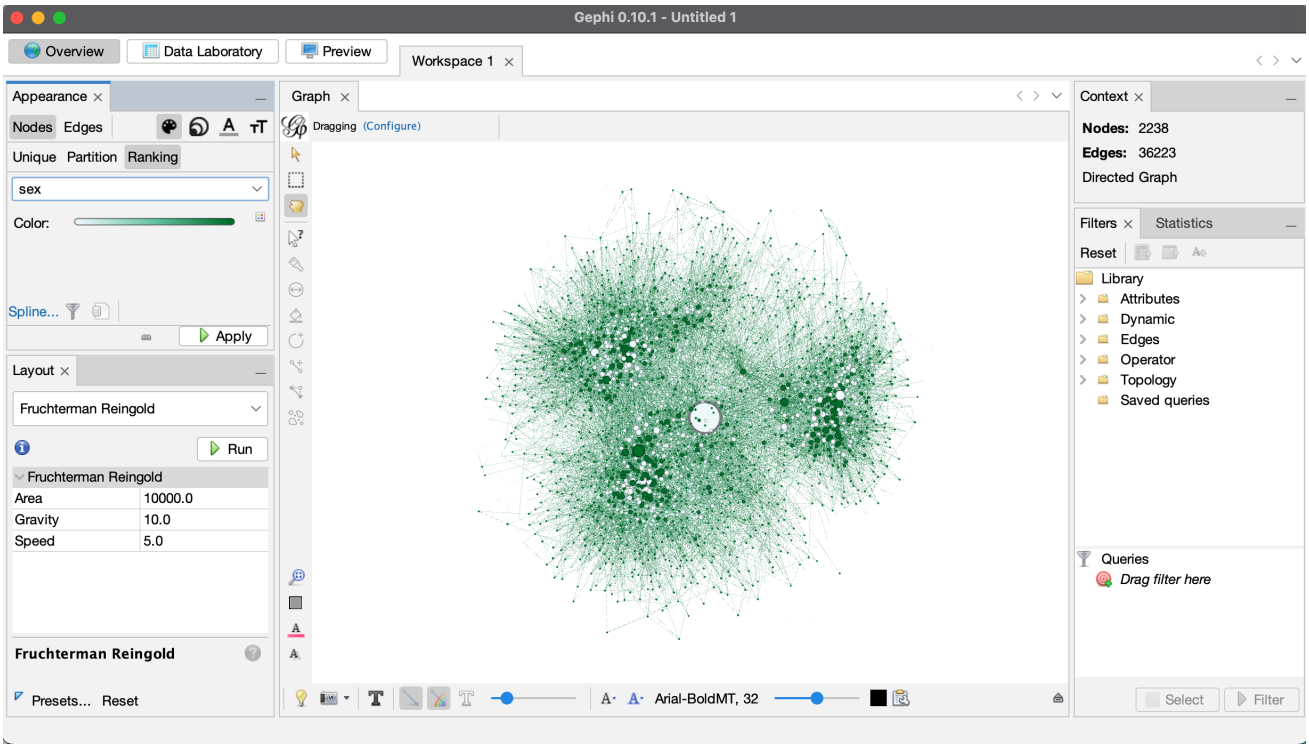
The initial network layout may appear messy or overlapping. To improve clarity, apply the **Fruchterman Reingold** layout algorithm:

- Go to the **Layout** panel (bottom left).
- Select **Fruchterman Reingold** from the dropdown menu.
- Adjust parameters such as **Area** or **Gravity** if needed (default settings usually work well).
- Click **Run** to start the layout process. You can increase the **Speed** to make the layout adjust faster.
- Watch as the nodes spread out and the structure becomes clearer. When the network stops moving, click **Stop** to finalize the layout.


Gephi 0.10.1 - Untitled 1
Overview
Data Laboratory
Preview
Workspace 1
Appearance
Nodes
Edges
Unique
Ranking
Degree
Min size:
10
Max size:
150
Spline...
Apply
Layout
Fruchterman Reingold
Run
Fruchterman Reingold
Area
10000.0
Gravity
10.0
Speed
5.0
Fruchterman Reingold
Presets...
Reset
Graph
Dragging
(Configure)
Center On Graph
Arial-BoldMT, 32
Context
Nodes: 2238
Edges: 36223
Filters
Statistics
Reset
Library
Attributes
Dynamic
Edges
Operator
Topology
Saved queries
Queries
Drag filter here
Select
Filter

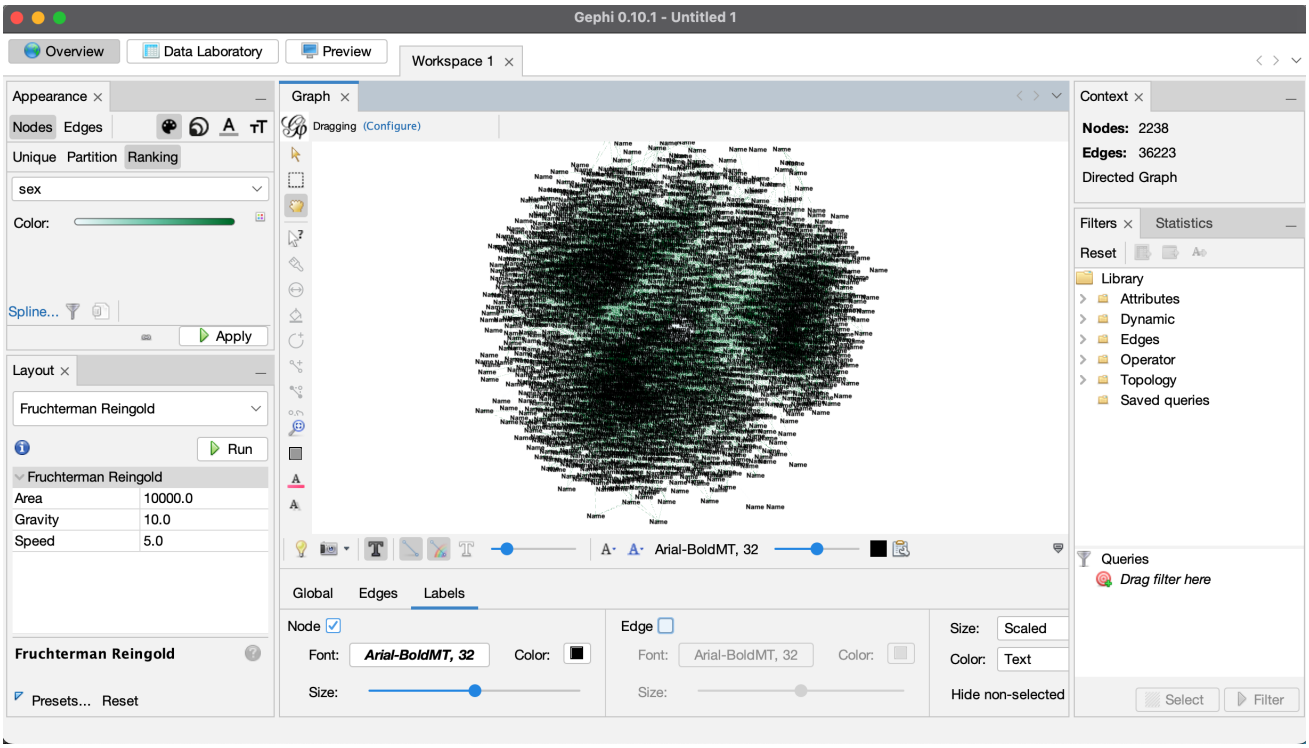
c. Color Nodes by Attribute

- In the **Appearance** panel, select **Nodes** and switch to the **Color** tab.
- Choose **Ranking** as the coloring method, then select **sex** from the dropdown menu. Click **Apply**.



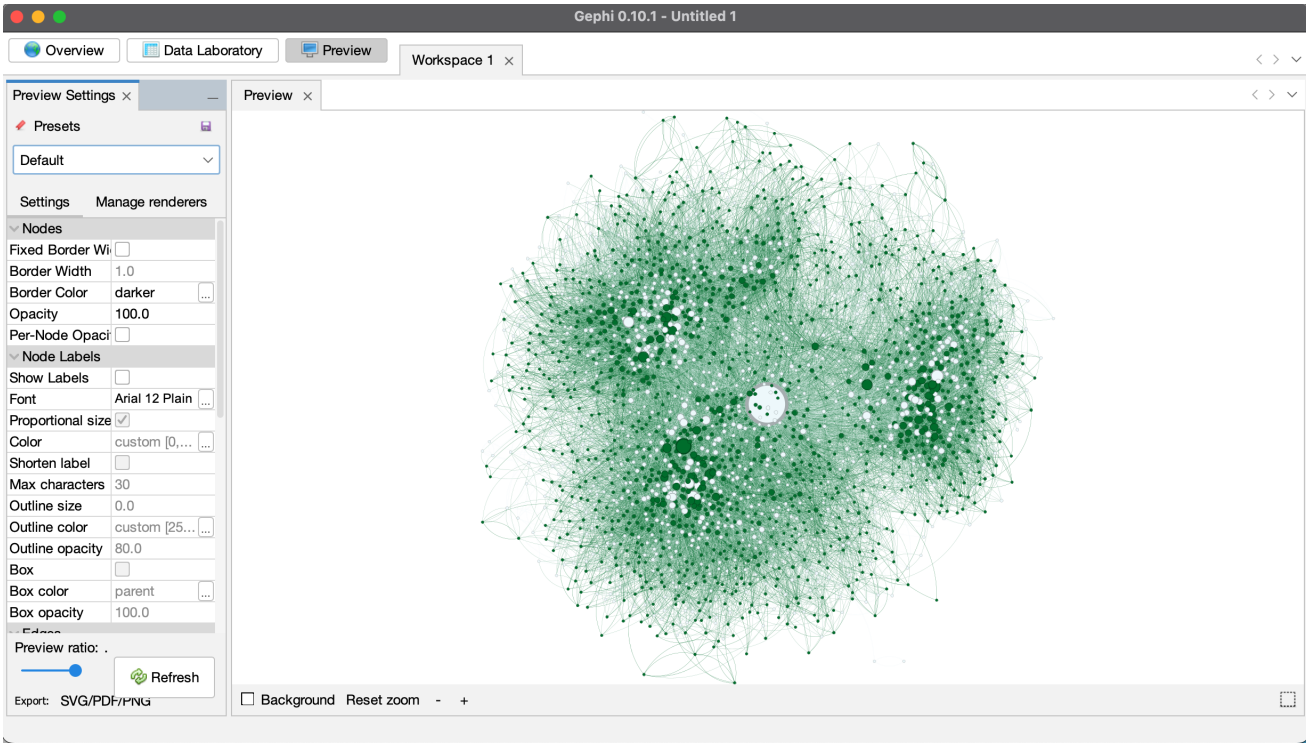
d. Show Node Labels

- Expand the toolbox (bottom right of the graph display). In the **Labels** panel, check **Node** to display labels.



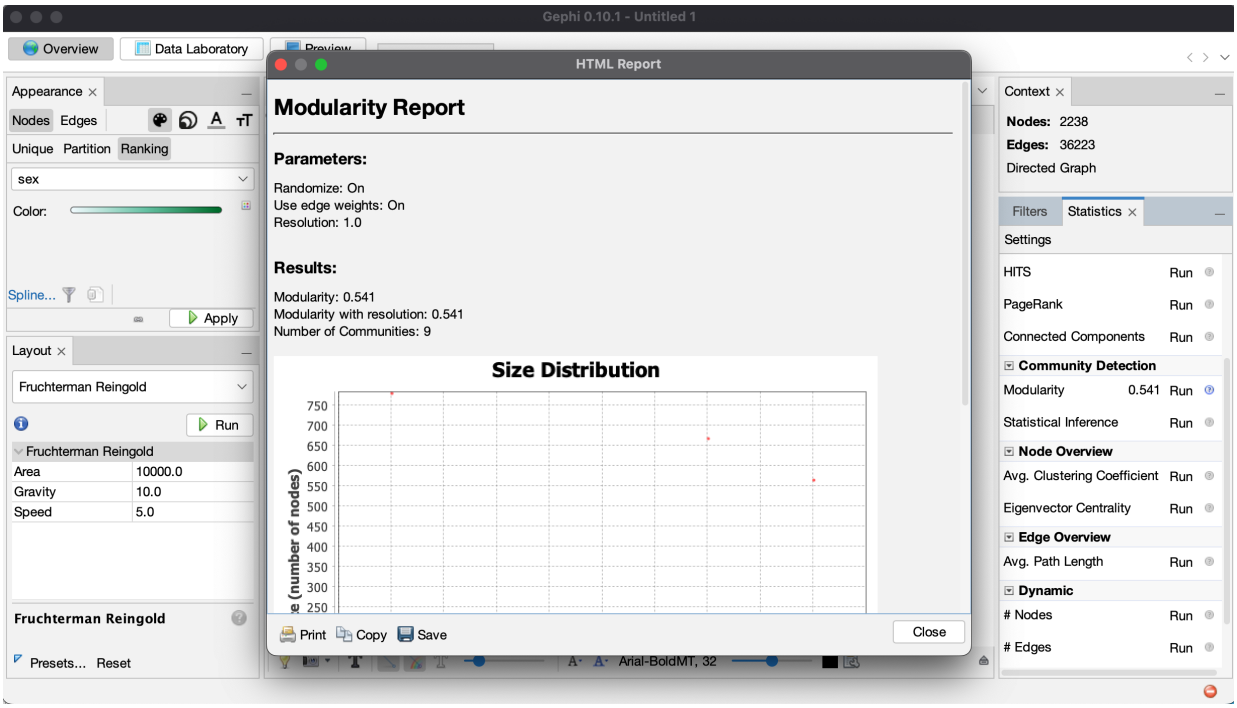
e. Final Touches and Export

- Go to the **Preview** tab for advanced rendering options. Adjust settings for better visuals (e.g., edge thickness, label size).
- Click **Refresh** to apply changes.
- **Export your visualization** in high quality by choosing **.svg** or **.pdf** formats.

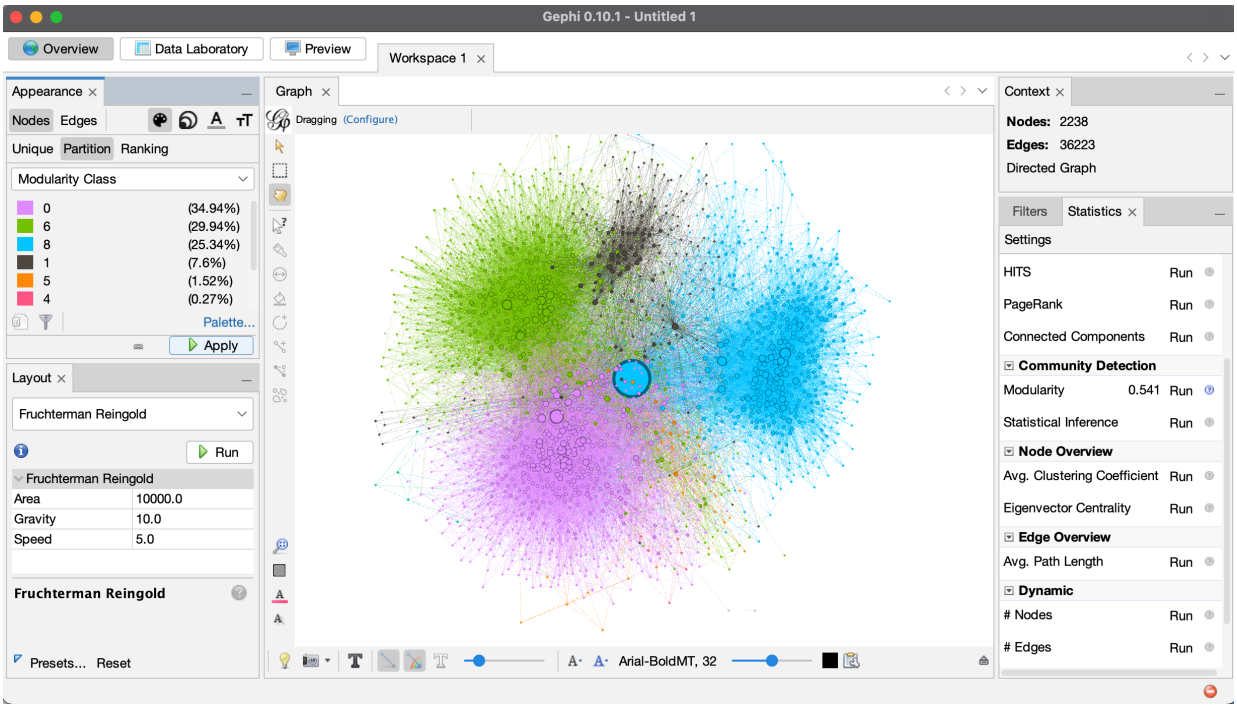


6. Network Analysis

- Use the **Statistics** panel (right bottom) to compute metrics like average degree, graph density, and modularity.
- To detect communities:
  - Run **Modularity** in the Statistics panel.
  - Choose a resolution (0.1–2), click **OK**.



- In the **Appearance** panel, color nodes by **Modularity Class** to highlight communities.



7. Your Task

1. Explore a different **Layout** algorithm (not the one you learned in lab) to rearrange the network visualization.
2. After applying the new layout, explore the **Filters** panel (right bottom). Under **Topology**, apply the **Degree Range** filter to remove some nodes from your network.
3. **Submission requirements:**
   - Take clear screenshots of your network **before filtering** and **after filtering**, both in a **new layout**.
   - Make sure the **entire Gephi interface** is captured.
   - Rename the screenshot of the original graph (before filtering) to `original`. Rename the filtered graph (after filtering) to `filtered`.
   - Both screenshots must clearly show the new layout you used and the degree range settings. Screenshots that do not clearly display both the layout and degree range settings will result in a deduction of marks.